

Localizing Factual Inconsistencies in Attributable Text Generation

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Abstract

There has been an increasing interest in detecting hallucinations in model-generated texts, both manually and automatically, at varying levels of granularity. However, most existing methods fail to precisely pinpoint the errors. In this work, we introduce QASEMCONSISTENCY, a new formalism for *localizing* factual inconsistencies in attributable text generation, at a fine-grained level. Drawing inspiration from Neo-Davidsonian formal semantics, we propose decomposing the generated text into minimal predicate-argument level propositions, expressed as simple question-answer (QA) pairs, and assess whether each individual QA pair is supported by a trusted reference text. As each QA pair corresponds to a *single* semantic relation between a predicate and an argument, QASEMCONSISTENCY effectively localizes the unsupported information. We first demonstrate the effectiveness of the QASEMCONSISTENCY methodology for human annotation, by collecting crowdsourced annotations of granular consistency errors, while achieving a substantial inter-annotator agreement. This benchmark includes more than 3K instances spanning various tasks of attributable text generation. We also show that QASEMCONSISTENCY yields factual consistency scores that correlate well with human judgments. Finally, we implement several methods for automatically detecting localized factual inconsistencies, with both supervised entailment models and LLMs.¹

1 Introduction

Large Language Models (LLMs) are used very effectively across a broad range of text generation tasks. However, despite remarkable progress in recent years, LLMs remain prone to generating factual inconsistencies. This phenomenon, commonly referred to as “hallucinations”, limits

their broader deployment and utility (Huang et al., 2023).

This work focuses on *attributable* text generation, where the generated content can be verified against a trusted supporting source, referred here as the “*reference text*”. This reference text may be part of the input for generation, as typical in text summarization or open book QA (Gao et al., 2023b; Slobodkin et al., 2024), retrieved post-generation (Bohnet et al., 2022; Gao et al., 2023a; Min et al., 2023; Wei et al., 2024), or identified by the models themselves when instructed to provide citations to external sources (Liu et al., 2023a; Yue et al., 2023).

To address hallucinations, there has been increasing research interest in identifying unsupported content in model-generated text, both manually and automatically. This task is typically framed as a textual entailment problem, requiring that the generated text should be *supported* (entailed) by the reference text. Detecting unsupported information is valuable for multiple purposes. For evaluating the factual consistency of attributable text generation, both human annotation protocols and automated inconsistency detection models are needed (Honovich et al., 2022; Gekhman et al., 2023). Automated inconsistency detection can further provide valuable feedback for end users about suspected unsupported content in the LLMs’ output, and also contributes to model improvements, via post-editing (Gao et al., 2023a), enhanced training (Nan et al., 2021; Wan and Bansal, 2022; Roit et al., 2023), self-critique (Wadhwa et al., 2024), or by imposing constraints during decoding (Wan et al., 2023).

To better fulfill these goals, it is desired to pinpoint *which parts* of the generated text are not supported, especially as LLMs continue to improve such that factual inconsistencies become more localized, as illustrated in Figure 1. While recent research has made useful strides

¹Our codebase, dataset, and models can be found at <https://github.com/ariecattan/qasemconsistency>.

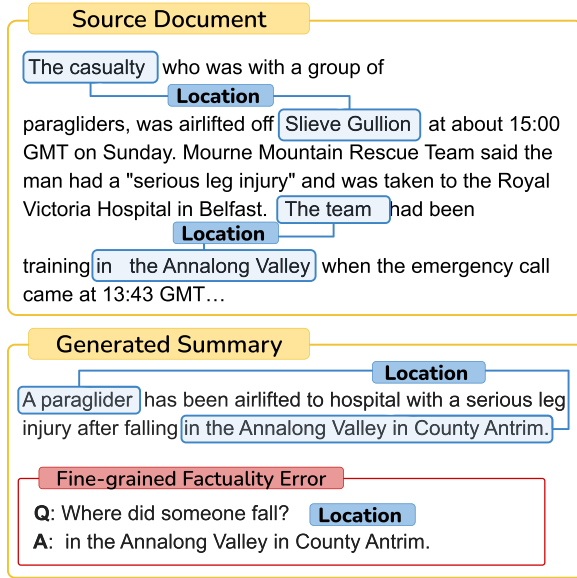


Figure 1: An annotation example of a localized factual consistency error according to our QASEMCONSISTENCY methodology. Here, the model successfully inferred a paraglider fall (after being airlifted with a serious leg injury) but incorrectly identifies the location of the fall as the rescue team’s training area (Annalong Valley) instead of the correct location (Slieve Gullion), which is located 30 miles away. This misattribution is highlighted by the question-answer pair: “Where did someone fall? in the Annalong Valley in County Antrim”.

in finer-grained inconsistency detection, ranging from entire texts to sentences, claims, and even question-generation and question-answering based solutions, these sub-sentence representations often remain insufficiently granular. For example, an “atomic” claim in FActScore (Min et al., 2023) is still based on multiple predicate-argument relations, and each of them might be either supported or unsupported (see Table 1, §2.1, and §6).

In this work, we introduce QASEMCONSISTENCY, a novel protocol for detecting localized factual inconsistencies in attributable text generation, applicable to both human and automatic detection. Inspired by Neo-Davidsonian formal semantics, our method decomposes the generated text into elementary assertions, in the form of atomic question-answer (QA) pairs (QA-SRL [He et al., 2015] and QANom [Klein et al., 2020]), where each pair corresponds to a single predicate-argument relation. Localizing factual inconsistencies then involves identifying the set of QA pairs that are not supported by the reference text. Figure 1 illustrates our QASEMCONSISTENCY

methodology by representing the factual inconsistency via a simple QA “Where did someone fall? in the Annalong Valley in County Antrim”, which is not supported by the reference text. By assessing each individual predicate-argument level statement, QASEMCONSISTENCY can pinpoint more precisely to the factual mistakes than prior methods (illustrated in Table 1). Notably, we found that for 27% of the predicates, some QA pairs were supported and some were not, justifying the need for such a fine-grained representation.

Furthermore, since we represent predicate-argument relations with simple natural language expressions (questions and answers), our QASEMCONSISTENCY methodology is well-suited for manually annotating granular consistency errors. Indeed, we collect a dataset with localized annotations of factual inconsistency at the predicate-argument level via crowdsourcing and achieve a high inter-annotator agreement (§4). We demonstrate that the overall factual consistency scores obtained by QASEMCONSISTENCY correlate well with human preferences.

Finally, we implement methods for automatically detecting whether each individual QASem QA is supported by the reference text, following the QASEMCONSISTENCY methodology (§5). We conduct experiments with a variety of models, including supervised NLI models and prompting open-source and commercial LLMs. While these models were supervised on standard entailment datasets, our results show that they can effectively handle our fine-grained QA assertions, providing more detailed error detection. Yet, there is a vast room for improvement in future work.

Altogether, we hope that future research will build upon QASEMCONSISTENCY for localizing factual inconsistencies (either manually or automatically) and leverage our benchmark to develop better entailment models for handling fine-grained hypotheses corresponding to predicate-argument relations.

2 Background

2.1 Fine-grained Detection of Factual Inconsistencies

Current efforts in detecting factual consistency errors continuously progress towards more localized methods that pinpoint the inconsistent information. Starting from approaches that highlight individual sentences (Laban et al.,

Source	Gareth Colfer-Williams, 25, died last week at his home in Swansea, the city at the centre of an epidemic of the disease which has reached 942 cases. But the examination was unable to establish whether measles was the main cause of his death. An inquest will be opened and adjourned on Tuesday to allow further tests... Public Health Wales said on Friday that laboratory tests confirmed a diagnosis of measles but further tests were needed to determine the cause of death...
Summary	An inquest into the death of a man who died of measles has been opened and adjourned after a post-mortem examination failed to establish how he got the illness. ✗
Token-level (CLIFF)	An inquest into the death of a man who died of measles has been opened and adjourned after a post-mortem examination failed to establish how he got the illness .
QGQA (Q^2)	- An inquest: What has been opened and adjourned into the death of a man who died of measles? ✓ - A man: Who died of measles? ✗ - measles: What was the cause of death of a man? ✗ - a post-mortem examination: What failed to establish how he contracted measles? ✗ - the illness: A post-mortem examination failed to establish how he got what? ✗
Claim-level (FActScore)	- An inquest into the death of a man has been opened. ✗ - The man died of measles. ✗ - A post-mortem examination was conducted. ✓ - The post-mortem examination failed to establish how he got measles. ✗ - The inquest has been adjourned. ✗
QASEMCONSISTENCY	<u>died/death:</u> - Who died? A man ✓ - How someone died? from measles ✗ <u>failed:</u> - What failed to do something? a post-mortem examination ✓ - What did something fail to do? to establish how he got the illness ✗ <u>got:</u> - Who got something? he; A man ✓ - What did someone get? the illness; measles ✓ <u>opened:</u> - What has been opened? An inquest into the death of a man ✗ <u>establish:</u> - What didn't establish something? a post-mortem examination ✓ - What didn't something establish? how he got the illness ✗ <u>examination:</u> - Who was examined? A man who died of measles ✓ - When was someone examined? post-mortem ✓ - Why was someone examined? to establish how he got the illness ✗ ...

Table 1: Comparison of QASEMCONSISTENCY to existing fine-grained decompositions. Here, the summary is inconsistent in multiple respects: It assumes that the man died from measles, while the cause of death remains unclear; the examination actually failed to establish the cause of death rather than how the man got the illness; and an inquest *will* be opened in the future rather than in the past. The token-level annotations are from CLIFF (Cao and Wang, 2021), with **highlighted** tokens indicating those marked as expressing unsupported information. For QGQA and claim-level, ✓ indicates “supported” and ✗ indicates “not supported”. QGQA lists the candidate answers (noun phrases) and the generated questions using Q^2 (Honovich et al., 2021), where the consistency labels indicate whether the candidate answer from the summary corresponds to the predicted answer based on the source. FActScore claims are generated by GPT 4o given FActScore’s instructions and demonstrations (Min et al., 2023), with our annotation of factual consistency. Unlike the other methods, our QASEMCONSISTENCY methodology decomposes the summary into fine-grained predicate-argument statements in the form of QAs, pinpointing more precisely at the unsupported facts. For example the unsupported QA “*How someone died? from measles*” pinpoints the inconsistency about the cause of death, while FActScore mixes in the same claim both the cause of death as well as the indication of who died, where the latter *is* consistent in the summary.

2022), inspect “facts” (Min et al., 2023), or use question-generation and question-answering (Honovich et al., 2021; Fabbri et al., 2022), researchers have developed methods to decompose the information in the generated text. While these decomposition techniques result in intuitive

propositions in natural language, they often lack granularity and include multiple units of information, each of which might be supported or not by the reference text. As exemplified in Table 1 (Claim-level), the extracted claim “the man died of measles” could be further decomposed into

two semantic relations - (1) “the man died” (PATIENT) and (2) “the death was due to measles” (CAUSE), with only the latter being unsupported by the reference text. We provide a detailed review of the relevant literature in Section 6.

Conversely, detecting factual inconsistencies at a finer level of granularity has involved using complex syntactic (Goyal and Durrett, 2020) and semantic (Ribeiro et al., 2022) formalisms. These formal approaches require linguistic expertise from annotators to manually evaluate generative models. Consequently, these methods were applied only for automatic detection of factual inconsistencies with models trained on synthetic data.

In this work, we represent minimal propositions, which correspond to individual predicate-argument relations, using intuitively comprehensible natural language question-answer pairs. This enables both human and automatic detection and a fine-grained representation of factual consistency errors.

2.2 Predicate-argument Relations: From Neo-Davidsonian Semantics to QA-SRL

Neo-Davidsonian semantics (Davidson, 1967; Higginbotham, 1983; Parsons, 1990) is a framework for representing the complex interactions between events and participants (i.e., predicate-argument relations) in a logical form. For example, given the sentence “*Mary bought a book yesterday and gave it to John with a smile*”, the Neo-Davidsonian representation is:

$$\begin{aligned} \exists e_1, e_2 & (\text{Buying}(e_1) \\ & \wedge \text{Agent}(e_1, \text{Mary}) \\ & \wedge \text{Theme}(e_1, \text{Book}) \\ & \wedge \text{Time}(e_1, \text{Yesterday}) \\ & \wedge \text{Giving}(e_2) \wedge \text{Agent}(e_2, \text{Mary}) \\ & \wedge \text{Recipient}(e_2, \text{John}) \\ & \wedge \text{Theme}(e_2, \text{Book}) \\ & \wedge \text{Manner}(e_2, \text{Smile})) \end{aligned} \quad (1)$$

This representation indicates that there are two predicates: *buy* (e_1) and *give* (e_2), each with its own set of arguments.

Following the underlying principles of Neo-Davidsonian semantics, various approaches have been developed to model fine-grained propositions corresponding to individual predicate-argument relations, including FrameNet (Baker et al., 1998), PropBank (Palmer et al., 2005),

bought	Who bought something? Mary
	What did someone buy? A book
	When did someone buy something? Yesterday
gave	Who gave something? Mary
	Who gave something to? John
	What did someone give? A book
	How did someone give? With a smile

Table 2: QA-SRL representation for “*Mary bought a book yesterday and gave it to John with a smile.*”

Semantic Dependency Parsing (Oepen et al., 2014), and AMR (Banarescu et al., 2013). These approaches, however, typically rely on complex semantic formalisms, making them less accessible to non-expert annotators, and in a sense harder to extract and manipulate with LLMs.

To bridge this gap, we propose using QA-SRL, a semantic formalism that simplifies traditional semantic role labeling (SRL) schemes by representing each predicate-argument relation through a simple and “minimal” question-answer pair, such as “*Who bought something? Mary*” (He et al., 2015; FitzGerald et al., 2018; Roit et al., 2020). For example, the QA-SRL representation of the above sentence is detailed in Table 2. Here, “*Mary*” is identified as the agent of the predicate “*bought*”, where this semantic relation is represented by the question “*Who bought something? Mary*”. Loosely speaking, each QA typically corresponds to a single Neo-Davidsonian proposition.

By relying on a comprehensible natural-language representation, QA-SRL largely subsumes traditional SRL schemes. Notably, it successfully covers valuable implicit semantic arguments (Roit et al., 2024), which are intuitively captured by human annotators, and subsequently by models trained on such annotated data.

3 QASEMCONSISTENCY

Given a generated text y and a reference text x that is expected to support the information in y (e.g., the source for generation or a grounding text for it), we define the task of localizing factual inconsistencies as identifying the set of “localized” assertions that are contained in y but are

not supported by x . This involves the decomposition of y into such *assertions*, each being a unit of information that can be individually assessed for its entailment by x . For effective localization, we suggest two desired properties of this decomposition. First, the assertions should be as *minimal* in scope as possible, where ideally each assertion should not be further decomposable into smaller verifiable assertions. Second, each assertion should be human interpretable, providing clear insights to common language speakers, thus allowing efficient crowdsourced annotation for localized factual inconsistencies.

To fulfill these properties, we propose decomposing y into the set of its predicate-argument level propositions, using the QASem framework (§2.2). By construction, each question-answer pair (QA) in QASem corresponds to a *single* predicate-argument relation, expressed in natural language.

A QA pair is considered supported by the reference text if the proposition corresponding to the predicate-argument relation is *entailed* from the reference text x . We follow previous work and frame the task as a binary classification problem, with labels $\in \{\textit{supported}, \textit{not supported}\}$, considering both the *neutral* and *contradiction* classes in the standard entailment recognition task as *not supported* (Maynez et al., 2020; Kryscinski et al., 2020; Honovich et al., 2022; Min et al., 2023). For example, consider the source document x and the summary y in Table 1, taken from our annotated dataset. The QA “Who died? A man” is *supported* because the article explicitly mentions that Gareth Colfer-Williams, who is a man, died. Conversely, the QA “How someone died? From measles” is *not supported*, as the article states that the cause of death remains unclear.

Once all QAs are assigned with a label, we can also calculate an overall factual consistency score, defined as the percentage of supported QAs over all QAs. This interpretable score represents the proportion of supported semantic relations within y and can be used for model evaluation. For example, the overall factual consistency score of the generated text in Table 1 is 7/12.² For comparison, the prior localization methods of QGQA and FActScore assign a score of 1/5 to the same summary. This discrepancy stems from the insufficient

granularity level of these approaches (see §2.1 and §6), where each assertion contains a mixture of supported and unsupported information.

QASem Parsing. We automatically generate QAs for both verbal and nominal predicates using a parser that we purposely trained on the QASRL (FitzGerald et al., 2018) and QANom (Klein et al., 2020) datasets. Specifically, we use the same architecture of the original QASem parser (Klein et al., 2022) and replace the original T5-small model with T5-XL (3B) to improve performance. This parser takes as input a sentence and a predicate, verbal or nominal, and generates a list of atomic QAs. We train our parser for 5 epochs until convergence with the Adam optimizer and a learning rate of $5e - 05$. We evaluate our parser on the QASRL gold data (Roit et al., 2020) and QANom (Klein et al., 2020), achieving 75.9 F1 (+7.3) on QASRL and 72.4 F1 (+13.2) on QANom. Similarly to Klein et al. (2022), we use a span match threshold of $\text{IOU} \geq 0.3$ to match between predicted and gold arguments. The above training datasets do not include annotation of predicate arguments for copular verbs (e.g., “John is a musician”). To ensure completeness of QASEMCONSISTENCY, we prompt Gemini-Flash (2.0) to generate QAs to represent predicate-argument relations for copular verbs (e.g., “Who is a musician? John”).

4 Human Detection of Localized Factual Inconsistencies

In this section, we apply our QASEMCONSISTENCY methodology to manually annotate localized factual inconsistencies in generated texts. We collect such annotations across three different scenarios of attributable text generation: summarization (Cao and Wang, 2021), generation of people’s biography and verification against their Wikipedia pages (Min et al., 2023), and response generation with a generative search engine citing external sources (Liu et al., 2023a).

This annotation serves two primary purposes. First, we assess whether QASEMCONSISTENCY is a suitable approach for collecting high-quality annotations of localized inconsistencies through cost-effective crowdsourcing. This can enable future work to perform human evaluation of generative models with this methodology. Second, we create a diverse and large entailment benchmark

²In practice, there are a few more QAs that we omitted for brevity.

where the hypothesis is a predicate-argument relation in the form of a question-answer pair. We believe this benchmark will be valuable for future research to develop and improve models that can predict entailment for predicate-argument level assertions. To the best of our knowledge, this is the first work to annotate factual inconsistencies for predicate-argument level propositions.

4.1 Data Collection

As mentioned in Section 3, given a generated text y , we automatically predict QA pairs that represent localized propositions corresponding to single predicate-argument relations, using our QASem parser.³ Then, human annotators inspect these QAs along with the reference text x and the generated text y and determine for each QA whether it is supported by x or not. Table 1 (QASEMCONSISTENCY) shows an example of such annotations. We now describe our complete annotation process.

Enhancing Annotation Efficiency. An entity mentioned in y might be absent in the reference text x (i.e., an extrinsic hallucination) (Xiao and Carenini, 2023). In such cases, all QAs featuring this entity as the answer are inevitably not supported. For instance, considering the source article in Table 1, if the summary would have mentioned “An inquest into the death of a **woman**...”, then the QAs “Who died? A woman” and “Who got something? A woman” would not be supported since the reference text does not mention any woman.

Leveraging this observation, we make the annotation process more efficient by dividing it into two sequential steps. In the first step, annotators go through the entity spans from the generated text y corresponding to QASem *arguments* (i.e., the answers) and classify each span to either “covered” or “not covered”, according to whether it is mentioned explicitly or can be implied from x . Any answer classified as “not covered” (representing an extrinsic hallucination) automatically renders all associated QA pairs as “unsupported”. This eliminates the need for annotators to evaluate these QA pairs individually.

In the second step, annotators focus exclusively on the remaining QA pairs—those whose answers

have been confirmed to be covered by the reference text x . We define a QA as “supported” if the meaning of that QA can be inferred from x . Specifically, we adhere to the original definition of textual entailment from (Dagan et al., 2013): “*a text T entails a hypothesis H if there exists some background knowledge K such that T and K together entails H while K alone does not*”. For instance, the reference text “Max was seriously injured when boiling water accidentally spilled on his hand” entails the response “Hot water over 80 degrees Celsius spilled on Max’s hand”, based *also* on the assumed common knowledge K that “Water boils at 100 degrees Celsius at sea level” but does not entail “Water reaches its boiling point at 100 degrees Celsius”, because the background knowledge K alone suffices to entail the text without the reference text. Therefore, annotators were instructed to rely solely on the reference text x and their common knowledge background to determine whether a QA can be inferred from the reference text. The use of external resources (e.g., web search) was restricted for clarifying the definitions of complex terms (e.g., “six-under-par”), but not to verify content not stated in the reference text.

To further assist annotators with QA evaluation, we advise them to rephrase the question-answer pair as an affirmative statement and assess whether the reference text supports this statement. For instance, the QA pair “*What did someone open? An investigation*” could be rephrased as “*Someone opened an investigation*”.

To enhance annotators’ focus, all QAs of the same predicate are shown together. In addition to the “support” labels, annotators were encouraged to write free text notes to justify their decisions, encouraging deeper considerations. These two separate stages improve annotation efficiency and also introduce an additional layer that classifies factual inconsistencies into extrinsic versus intrinsic errors.

Annotation Tool. To facilitate the human annotation process, we develop an intuitive annotation interface that streamlines the two steps.⁴ Figure 2 shows the interface of the second annotation step (QA evaluation). See Appendix A for implementation details.

³In some cases, low-quality generated QAs were filtered in the annotation process, in a preliminary step (see Appendix A).

⁴Our tool can be found at <https://github.com/ariecattan/loc-unfaith>.

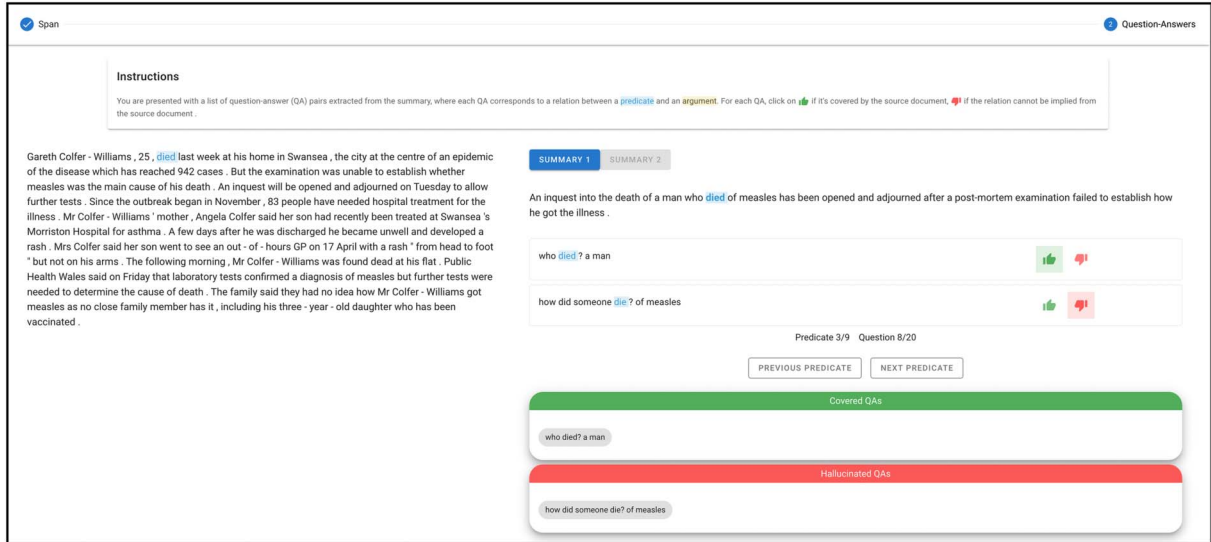


Figure 2: An example of the QA annotation step. The article in the left side is the reference text and the summary is shown on the right side. At each time, the interface highlights a specific predicate (here “died”) and displays the QAs representing the predicate-argument relations for that predicate. The green thumbs-up and red thumbs-down correspond to supported and not supported, respectively.

Tasks and Generative Models. We collect human annotations from three settings of attributable text generation.

First, we consider the task of abstractive summarization on the XSUM dataset that summarizes news articles to a single sentence. Specifically, we sample a subset of 74 summaries from the CLIFF dataset (Cao and Wang, 2021), in which the source articles are from XSUM (Narayan et al., 2018) and the summaries were automatically generated by BART (Lewis et al., 2020) and PEGASUS (Zhang et al., 2020). CLIFF manually annotated each generated summary with token-level annotation of consistency errors.

Second, we annotate faithfulness localization for 34 people biographies (3–5 sentences) included in FActScore (Min et al., 2023). These biographies were generated in zero-shot by LLMs such as ChatGPT, InstructGPT, and the retrieval-augmented PerplexityAI model⁵ and were subsequently verified against the Wikipedia page of these entities.

Lastly, we extend our evaluation to open-ended response generation with in-line citations to external sources. We use the “Verifiability” dataset (Liu et al., 2023a) that assesses factual consistency of each generated sentence against its corresponding source(s) for several generative search engines (BingChat, NeevaAI, Perplexity.ai, and YouChat)

⁵perplexity.ai.

Task	#Responses	#Sentences	#QAs	IAA
CLIFF	74	74	693	0.72
FActScore	36	229	1,109	0.78
Verifiability	41	189	1,296	0.67

Table 3: Statistics of our collected benchmark.

across a range of diverse queries (AllSouls, davincidebate, WikiHowKeywords, ELI5 [KILT / Live], NaturalQuestions). From this dataset, we selected 41 responses, comprising a total of 189 sentences.

Each instance in our dataset includes annotations from 3 different workers.

Annotators. We recruit annotators through Amazon Mechanical Turk.⁶ We follow the controlled crowdsourcing protocol (Roit et al., 2020), which consists of training the workers with detailed instructions (using intuitive slides) and providing ongoing personalized feedback throughout the process. Multiple examples were included to illustrate the two annotation tasks (Span and QA evaluation). Annotators’ compensation is described in Appendix A.

4.2 Dataset Properties

Table 3 presents the statistics of our collected dataset. Overall, we gathered entailment annotations for 3,098 different QAs. The ground truth

⁶<https://www.mturk.com/>.

label for each QA is determined by the majority vote among the annotators. We split the dataset into development and test sets 50/50. The development set can serve for prompt engineering or for optimizing the decision threshold.

Each sentence is represented by an average of 6.3 predicate-argument QAs. This granular decomposition contrasts with FactScore’s approach, which yields only 4.1 free-text claims per sentence. In addition, we found that for 27% of the predicates in our collected benchmark, some QA pairs are entailed and some are not. This confirms that assessing each predicate-argument assertion in the model-generated response is valuable and allows us to pinpoint the factual inconsistencies.

We evaluate inter-annotator agreement using Fleiss’ Kappa. We observed substantial agreement across all three tasks: $\kappa = 0.72$ for CLIFF, $\kappa = 0.79$ for FactScore, and $\kappa = 0.67$ for Verifiability. For comparison, previous factuality benchmarks report lower agreement: Pagnoni et al. (2021) report $\kappa = 0.58$ at the sentence-level and Cao and Wang (2021) report $\kappa = 0.35$ at the token-level. Annotating factual consistency is a challenging and sometimes subjective task (Falke et al., 2019). We believe that the high agreement is due to the QASEMCONSISTENCY’s decomposition into atomic semantic relations, where annotators need to assess a single assertion at a time. In addition, by collecting factual consistency annotation for each predicate-argument relation, QASEMCONSISTENCY ensures that the human raters will assess all details.

4.3 Analysis

4.3.1 Overall Score

Here, we aim to show that QASEMCONSISTENCY not only enables the localization of factual inconsistency but also provides an overall score that reflects well the degree of inconsistency in the response. To achieve this, we collect side-by-side annotations of a source text x paired with two different model-generated responses y_1 and y_2 . Annotators were asked to compare the factual consistency of the two responses on a scale of 1 to 5, where 1 indicates that y_2 is *much* more consistent than y_1 , 2 that y_2 is more consistent than y_1 , 3 that they are almost equivalent and symmetrically 4 and 5 indicate advantage of y_1 over y_2 . This pairwise comparative judgment (Thurstone, 1927; David, 1963), a common and intuitive approach

for comparing model outputs, allows human annotators to reliably compare outputs on a shared scale (Chiang et al., 2024).

Given a scoring function $s(x, y)$ that assigns a factual consistency score to the output y with respect to the reference text x , we expect the difference $d(y_1, y_2) = s(x, y_1) - s(x, y_2)$ to correlate with the side-by-side score. Indeed, if y_1 is much more consistent than y_2 , we expect d to be a large positive value, whereas if the two responses are nearly equivalent in consistency, d should be near zero.

We collect these side-by-side annotations on the summaries from CLIFF (37 pairs) and the biographies from FactScore (16 pairs).⁷ As mentioned above (§3), we define $s_{QA}(x, y)$ as the percentage of supported QAs in y and compute this score using human labels. The Spearman correlation between $d(y_1, y_2)$ and the side-by-side consistency score yields strong positive results. For the summaries, we obtain a high correlation of $\rho = 0.73$ (p-value < 0.001), while the token-level evaluation in CLIFF ($\rho = 0.11$) and QGQA ($\rho = 0.2$) show no correlation. Figure 3 plots the difference in factual consistency scores derived from QASEMCONSISTENCY and from token-level annotations against human side-by-side preferences. For the biographies, we obtain a high correlation for all metrics ($\rho = 0.71$ for QASEMCONSISTENCY, $\rho = 0.67$ for FactScore, $\rho = 0.81$ for QGQA). However, due to the overlapping confidence intervals and the small sample size ($N = 16$), we cannot conclusively determine that one metric is statistically superior to the others based solely on these results. **These strong correlations demonstrate that QASEMCONSISTENCY effectively reflects the degree of inconsistency in model-generated responses and can be reliably used for ranking models.**

4.3.2 Qualitative Analysis

We compare the localized annotations of supported QAs in the CLIFF’s subset of our dataset to the original token-level annotations in CLIFF. Specifically, we analyze the cases where summaries were annotated as *fully supported* in CLIFF but not in our annotations and vice-versa. Notably, we observe that all ten summaries where all QAs were annotated as *supported* in our dataset were

⁷We cannot do this analysis for verifiability because different models generate responses that point to different sources.

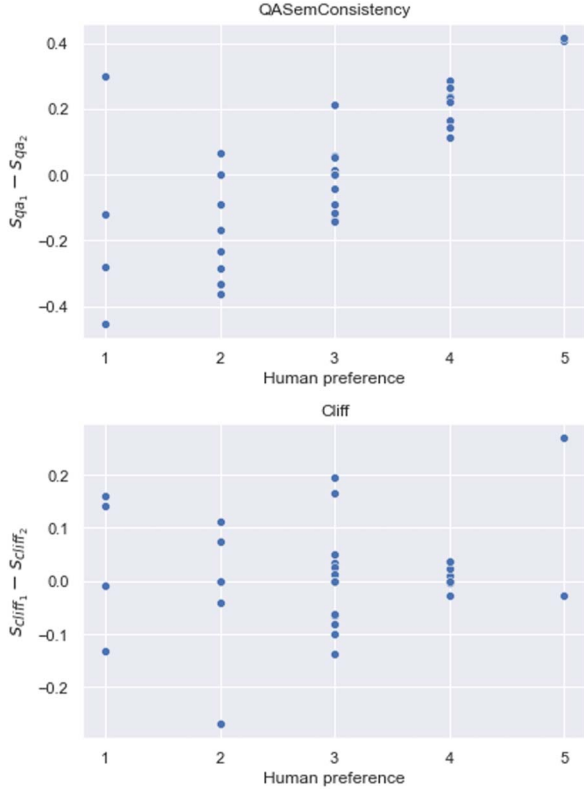


Figure 3: Visualization of the correlation between the difference in factual consistency scores (calculated using QASEMCONSISTENCY and token-based approach for “Cliff”) and human side-by-side preferences.

also annotated as *fully supported* in CLIFF. Conversely, we identified 10 out of 74 summaries marked as *fully supported* in CLIFF whereas our annotators found unsupported QAs. Upon analyzing these cases, we discover that the majority of the summaries (7 out of 10) were indeed not fully supported by the reference text, as reflected by the unsupported QAs. For instance, the summary sentence “*The US has suspended its participation in talks with Russia to try to broker a cessation of hostilities in Syria, the State Department says.*” is not fully supported by the reference text “*The United States is suspending its participation in bilateral channels with Russia that were established to sustain the cessation of hostilities*” because the reference text mentions that the suspension of the U.S participation is “*to sustain the cessation*” and not “*to broker a cessation*”. This fine-grained factual inconsistency was indicated by several QAs marked as *not supported*, such as “*Why has someone suspended something? To try to broker a cessation of hostilities in Syria*” about the predicate “*suspended*”. Two other summaries could be interpreted as either

supported or not supported, with our annotators showing localized disagreement on these cases. One summary was mistakenly annotated as not supported by our annotators. This analysis confirms that having human annotators verify each predicate-argument relation is not only beneficial for localizing factual inconsistencies, but also effectively helps annotators to *identify* them more accurately.

5 Automatic Detection of Localized Factual Inconsistencies

In this section we describe several models that automatically localize factual inconsistencies according to our methodology. We decompose the generated text y into a list of QA pairs $\{qa_1, qa_2, \dots, qa_n\}$, and define the likelihood for each qa_i to be supported (entailed) by the reference text x as $s(x, qa_i) \in [0, 1]$. We use the QASem parser from Section 3 to generate the QAs and conduct our experiments with different entailment classifiers.

5.1 Entailment Classifiers

We experiment with two types of models:

Supervised. We apply three recent off-the-shelf NLI models to predict whether the QASem QA qa_i is entailed by the reference text. In these experiments, the premise is the reference text x and the hypothesis is the concatenation of the question q_i and the answer a_i in qa_i .

The first supervised classifier is **TRUE** (Honovich et al., 2022), an encoder-decoder model based on T5-XXL (11B parameters) (Raffel et al., 2020) and finetuned on diverse entailment datasets: SNLI (Bowman et al., 2015), MNLI (Williams et al., 2018), FEVER (Thorne et al., 2018), Scitail (Khot et al., 2018), PAWS (Zhang et al., 2019), and VitaminC (Schuster et al., 2021).⁸ The TRUE model, also known as AUTOAIS (Bohnet et al., 2022), was widely used in previous work for automatically measuring factual consistency (Bohnet et al., 2022; Gao et al., 2023a; Roit et al., 2023; Gao et al., 2023b; Slobodkin et al., 2024). This model was trained to generate “1” if the hypothesis is entailed by the premise and “0” otherwise. Hence, $s(x, qa_i)$ is set as the probability of generating “1”, given the likelihood

⁸<https://huggingface.co/google/t5-xxl-true-nli-mixture>.

	CLIFF		FActScore		Generative Search	
	BAcc	AUC	BAcc	AUC	BAcc	AUC
<i>Supervised NLI models</i>						
TRUE (11B)	61.4	72.1	72.5	88.8	68.5	79.1
TrueTeacher (11B)	68.9	82.1	71.7	88.7	57.8	80.5
Minicheck (770M)	70.4	80.8	78.0	87.3	76.3	83.4
<i>LLM prompting</i>						
Gemma 2 2B	65.9	73.7	74.6	81.9	70.8	80.2
Gemma 2 9B	70.0	82.5	78.0	88.1	77.1	87.1
Gemma 2 27B	72.3	85.2	75.2	90.4	71.6	85.5
Mistral Nemo (12.2B)	70.6	79.5	73.7	85.4	71.5	82.4
Llama 3.1 8B	65.8	79.0	76.5	89.4	68.5	83.7
GPT-4o	75.7	—	82.4	—	79.2	—

Table 4: Performance of automatic models on the test set of our collected benchmark. We cannot report AUC for GPT-4o because it does not directly output probability scores for “Yes” or “No” answers. The highest BAcc and AUC within each category (Supervised NLI model and LLM prompting) are shown in **bold**.

score of “1” and “0”. The second supervised model is **TrueTeacher** (Gekhman et al., 2023), a T5-XXL model finetuned on many model generated summaries annotated for factual consistency using an LLM. Similarly to TRUE, this model was trained to generated “1” if the hypothesis is entailed by the premise. The last supervised model is **MiniCheck** (Tang et al., 2024), a recent T5 Large (770M parameters) trained on synthetic data generated with GPT-4. This model outperforms all systems of comparable size and reaches GPT-4 accuracy on LLM-AggreFact.

LLM Prompting. We also use several LLMs to predict $s(x, qa_i)$ using few-shot prompting. We instruct the model to write “Yes” if the proposition corresponding to the QA is entailed by the reference text, and “No” otherwise. The prompt used in our experiments is presented in Appendix C. We first apply this prompt to recent open-source LLMs including **Llama 8B (v3.1)** (Dubey et al., 2024), **Gemma (v2) 2B, 9B, and 27B** (Riviere et al., 2024), and **Mistral Nemo Instruct**.⁹ For all prompted models, we define the entailment score s as the probability of generating “Yes” as the first generated token, based on the likelihood score of generating “Yes” and “No”. Finally, we also use GPT-4o to predict entailment for the QAs but evaluate only “hard” predictions (see Section 5.2) because it does not output distributions over the vocabulary.

⁹<https://mistral.ai/news/mistral-nemo>.

5.2 Evaluation of Automatic Detection

We report two common metrics to evaluate the performance of automatic models to predict entailment of QAs. First, we follow standard evaluation practices (Honovich et al., 2022; Gekhman et al., 2023) and report the Area Under the Receiver Operating Characteristic Curve (ROC AUC), which plots the true positive rate against the false positive rate for different possible thresholds of $s(x, qa_i)$. Second, since the models are trained or instructed to classify a QA as either *supported* or *not supported*, we also measure the performance of the hard prediction. Similarly to Tang et al. (2024), we do not perform threshold tuning for each dataset and consider a QA as supported if $s(x, qa_i) \geq 0.5$. Following previous work (Laban et al., 2022; Fabbri et al., 2022; Tang et al., 2023, 2024), we report the Balanced accuracy metric (BAcc): $BAcc = \frac{1}{2}(\frac{TP}{TP+FN} + \frac{TN}{TN+FP})$.

5.3 Results

Table 4 presents the results of our automatic models. Notably, GPT-4o achieves the highest balanced accuracy (BAcc) across all datasets (75.7 on CLIFF, 82.4 on FactScore and 79.2 on Verifiability). Open source LLMs and supervised NLI models also achieve decent performance, with Minicheck emerging as the top-performing supervised model. These results suggest that even though these models were trained on standard entailment datasets, where the hypothesis

is a single sentence, they can effectively adapt to our setting, where hypotheses take the form of question-answer pairs representing predicate-argument relations (e.g., “*How someone died? from measles*”). This generalization ability likely stems from the models’ training on massive text corpora, coupled with the fact that QASem question-answer pairs are expressed in natural language, unlike traditional SRL schemes like PropBank or FrameNet that rely on predefined and complex taxonomies. Finally, the consistently higher performance on the FactScore dataset can be explained by the nature of its biographical content. These texts contain a high frequency of copular sentences (e.g., “*Roselyn Sanchez is an actress.*”), which present a simpler verification challenge for the models. Indeed, 15% of the QAs in FactScore are copular constructions (e.g., “*Who is an actress? Roselyn Sanchez*”, for which GPT-4o achieves a BAcc of 0.89, compared to 0.81 for non-copula QAs).

Beyond standard metrics, we want to assess how well the model’s overall scores align with human judgment, similar to our approach with human scores (see Section 4.3.1). We compute the Spearman correlation between the difference of the individual QA scores $d(y_1, y_2) = s_{QA}(x, y_2) - s_{QA}(x, y_1)$ and the side-by-side human preference Likert scale. Our experiments show that GPT-4o achieves a Spearman correlation of $\rho = 0.54$ (p-value < 0.05) on CLIFF.¹⁰

While GPT-4o achieves a high performance across datasets, there is still a gap between automatic and human performance, leaving much room for improvement in future work.

5.4 Analysis

QA vs. Affirmative. To investigate the impact of the question-answer (QA) format on entailment performance, we convert the QA pairs into affirmative sentences (e.g., “*Who ate something? John*” to “*John ate something*”) using a small LLM (Gemma 2 2B). We then prompt Gemma 2 9B to determine whether these affirmative sentences are supported by the reference text. Surprisingly, this approach results in substantially lower balanced accuracy (BAcc) scores compared to the QA format: 63.7 (−6.3) on CLIFF, 74.7 (−3.3) on FActScore, and 73.4 (−3.7) on Verifi-

ability. We believe that the QA format provides more structured information than simple affirmative sentences because (1) it explicitly delineates the predicate (within the question) and the argument (within the answer), and (2) the question formulation (e.g., “who”, “where”, etc.) offers an additional layer of semantic information which can be valuable for deciding entailment.

6 Related Work

Identifying factual inconsistencies in attributable text generation has become a prominent research area in recent years. Existing methodologies for this task can be categorized based on the granularity of the detection. Table 1 illustrates and compares the various decomposition methods, recently proposed in the literature.

Starting with coarse-grained evaluation, SummEval (Fabbri et al., 2021) asks annotators to assign a 1–5 Likert score to the entire summary, while some other works aim to produce a single score/label for the entire output (Yin et al., 2021; Rashkin et al., 2023; Honovich et al., 2022; Tang et al., 2023; Liu et al., 2023b; Gekhman et al., 2023). Several works evaluate each generated sentence separately with some (simple or sophisticated) form of aggregation (Falke et al., 2019; Kryscinski et al., 2020; Pagnoni et al., 2021; Utama et al., 2022; Tang et al., 2022; Laban et al., 2022; Mishra et al., 2024; Subbiah et al., 2024; Tang et al., 2024).

To achieve sub-sentence evaluation, a few recent works decompose each sentence into “claims”, “facts”, or “propositions” (Table 1 Claim-level), whose support by the reference text is then assessed independently (Min et al., 2023; Krishna et al., 2023; Chen et al., 2023; Kamoi et al., 2023a; Wanner et al., 2024; Samir et al., 2024; Wei et al., 2024; Wan et al., 2024). However, these free text claims typically lack structure and systematicity. As a result, identifying unsupported claims does not effectively *localize* factual mistakes in the generated text. In fact, these claims are not atomic and encompass multiple semantic relations. For instance, the claim “The man died of measles” in Table 1 is not supported by the article, but it remains unclear whether the issue is that the man died for a reason other than measles, or that there is no man who died.

Another line of automatic evaluation methodology, typically referred to as question-generation

¹⁰The results on FactScore were not statistically significant due to the limited number of pairwise preferences ($n = 16$).

and question-answering (QGQA), consists of generating questions and answers based on the model output and then comparing the answers to those obtained from the reference text by a QA model (Wang et al., 2020; Durmus et al., 2020; Nan et al., 2021; Scialom et al., 2021; Honovich et al., 2021; Fabbri et al., 2022). However, Kamoi et al. (2023b) demonstrate that this paradigm falls short in providing effective *localization* of factual inconsistency. This is primarily because the generated questions often contain factual inconsistencies from the summary itself. Indeed, as shown in Table 1 (QGQA), the generated question for the answer phrase “an inquest” assumes that the man died of measles and something was already opened and adjourned. In contrast, our method is based on predicate-argument level QAs, where each QA represents a *single* semantic relation.

Some other works ask human annotators to highlight inconsistent tokens or spans in the generated text (Maynez et al., 2020; Cao and Wang, 2021). This method often results in relatively poor inter-annotator agreement (e.g., 0.35 Fleiss Kappa in CLIFF), because span annotation is subjective (Mishra et al., 2024) and individual spans might serve multiple roles in the sentence, where some roles are in correct assertions and some are not. Indeed, the token-level evaluation in Table 1 includes the tokens “*how he got the illness*”, although this phrase also implies that the man got the illness, which is supported by the reference text. In that summary, the mistake is that the goal of the examination should be to determine the cause of death, rather than how the man contracted the illness. This unsupported fact cannot be captured with span highlighting while it can effectively identified using our QA “*What didn’t something establish? how he got the illness*” (Table 1). In the same line of work, Laban et al. (2023) create SUMMEDITS, a challenging benchmark with localized factual inconsistencies. However, unlike our benchmark, the mistakes are limited to a single token and are not naturally occurring.

Goyal and Durrett (2020) propose DAE, an automatic evaluation metric that operates at the level of semantic dependency arcs in a structured semantic dependency representation (Oepen et al., 2014) to localize inconsistent semantic relations in the generated text. Similarly, FactGraph (Ribeiro et al., 2022) represents both the source article and the summary with AMR (Banarescu

et al., 2013) then model the factual inconsistencies at the edge level. However, these methods have only been applied automatically because obtaining human judgments at this level of granularity is challenging. Indeed, annotators would need to understand dependency or AMR labels and isolate the semantics of individual arcs within sentences, making manual evaluation difficult. Similarly to DAE and FactGraph, our QASEMCONSISTENCY also operates at the level of individual semantic relations, while representing them with simple natural language question-answer pairs, enabling human evaluation as well and easing LLM processing. Cho et al. (2024) shown success at applying Neo-Davidsonian formal semantics to automatic text-to-image evaluation.

We note though that while QASEMCONSISTENCY is a promising approach for both manual and automatic evaluation, it focuses solely on measuring factual consistency, a crucial aspect of attributable text generation. To provide a more comprehensive assessment, we suggest that future work use QASEMCONSISTENCY *in addition* to targeted metrics for capturing other aspects, such as BERTScore (Zhang et al., 2020) for relevance, BookScore (Chang et al., 2024) for coherence, or with LLM as a judge (Zheng et al., 2023).

Furthermore, it is worth noting that recent advancements in reference-based metrics for the relevance aspect, such as Pyramid (Nenkova and Passonneau, 2004), LitePyramid (Shapira et al., 2019), and RoSE (Liu et al., 2023c), highlight the growing interest in more granular evaluation of relevance. Since QASEMCONSISTENCY decomposes text into fine-grained predicate-argument assertions, we conjecture that it could be adapted also for providing more fine-grained evaluation of relevance.

7 Limitations

Our work has several limitations. First, as we consider only verbal and nominal predicates, factual inconsistencies that stem from other predication types, e.g., adjectives, will not be localized with the finest granularity. For example, consider the reference text “*John needs to repair his new red car, after the accident*”, and the generated text “*John’s blue car was damaged after he made an accident*”. QASEMCONSISTENCY would identify the inconsistency with the QA “*What was damaged? John’s blue car*”. However, this QA

could be further divided into two smaller QAs “*What was damaged? John’s car*” and “*What is blue? John’s car*”, while only the latter is not supported. In addition, similarly to previous decomposition approaches (Min et al., 2023; Krishna et al., 2023), QASEMCONSISTENCY does not capture factual inconsistencies that are due to implicit or inter-sentence discourse relations. For example, the sentences “*John missed the train and arrived late*” and “*John arrived late and missed the train*” would be treated as equivalent, even though they describe opposite causes and consequences. These limitations could be potentially addressed in future work by enriching QASEMCONSISTENCY with additional semantic decompositions, such as QADiscourse (Pyatkin et al., 2020) or QA-Adj (Pesahov et al., 2023), when improved parsers will be made available.

8 Conclusion

We introduced QASEMCONSISTENCY, a methodology that detects and localizes factual consistency errors to specific predicate-argument relations. Our error localization method is robust, as shown by our high human agreement and strong correlation with human preferences. Contrasted with other methods that either created complex, non-granular claims in natural language, or relied on linguistic formalisms that excluded non-expert annotators, our method can be applied with ease by non-expert users and models alike. Moreover, it can help human consumers of generative models recognize which parts of a response should be double checked.

We hope that future work will focus on extending our approach to detect inconsistencies in the wider discourse, and help generative models to correct their responses.

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A Dataset Collection

Filtering Nonsensical QAs Although our QASem parser significantly outperforms the current SOTA model (Klein et al., 2022) by a large margin (see §3), it is still prone to errors that can introduce noise into the consistency evaluation. Specifically, for nominal predicates, the parser may generate QAs that are nonsensical or convey a different meaning from the intended predicate. For instance, in the sentence “Sweden’s former foreign minister Johan Gustafsson, who was kidnapped by Islamist militants in Mali in 2013, has been released after more than two years in captivity” (from CLIFF) the parser incorrectly generates “Where did someone *militantise*?” for the predicate “militants” and “Who was *captained*?” for the predicate “captivity”. In some rare cases, the parser hallucinates predicate argument relation. For instance, in the sentence “A man has been arrested on suspicion of murder after a man was found dead at a house in Cambridgeshire”, the parser generates the question “Where was someone arrested? at a house in Cambridgeshire”, while this location refers to the predicate “found”. To mitigate this issue, we add a preliminary annotation step, in which an annotator was asked to verify whether the QAs are both (1) semantically interpretable

Span Question-Answers

Instructions
Your goal is to determine whether the entity spans in the summary below are correct or hallucinated. That is, for each **span**, click on **+** if it's covered by the source document (left) and on **-** if it's an hallucination.

Gareth Colfer - Williams, 25, died last week at his home in Swansea, the city at the centre of an epidemic of the disease which has reached 942 cases. But the examination was unable to establish whether measles was the main cause of his death. An inquest will be opened and adjourned on Tuesday to allow further tests. Since the outbreak began in November, 83 people have needed hospital treatment for the **illness**. Mr Colfer - Williams' mother, Angela Colfer said her son had recently been treated at Swansea's Morriston Hospital for asthma. A few days after he was discharged he became unwell and developed a rash. Mrs Colfer said her son went to see an out-of-hours GP on 17 April with a rash "from head to foot" but not on his arms. The following morning, Mr Colfer - Williams was found dead at his flat. Public Health Wales said on Friday that laboratory tests confirmed a diagnosis of measles but further tests were needed to determine the cause of death. The family said they had no idea how Mr Colfer - Williams got measles as no close family member has it, including his three-year-old daughter who has been vaccinated.

SUMMARY 1 **SUMMARY 2**

An inquest into the death of a man who died of measles has been opened and adjourned after a post-mortem examination failed to establish how he got the **illness**.

Covered Spans **Hallucinated Spans**

man measles post-mortem he illness

Notes (optional)

NEXT STEP

Figure 4: An example of the first annotation step.

and (2) correctly represent the semantic relations in the generated text. This preliminary step is efficient and required less than an hour to validate 500 QAs.

Annotation Interface Figure 4 shows the interface of the first annotation step (entity evaluation) and Figure 2 shows the interface of the second annotation step (QA verification). Once annotators finish the first step, they can move the second step, but they can always come back and modify their annotation for the entity evaluation.

Our annotation interface was developed with Vuejs¹¹ as a WebComponent, which can embed into any website and popular annotation platforms, such as Amazon Mechanical Turk.

Annotator Compensation For XSUM and biographies, each HIT consists of one or reference text, $\mathcal{X}$, and two model outputs, $\mathcal{Y}_1$ and $\mathcal{Y}_2$. For XSUM, each HIT includes a source article with summaries generated by BART and PEGASUS, and compensated with \$1.50 per HIT. For biographies, we select two model outputs for each instance, with a compensation of \$3.00 per HIT, reflecting the larger number of QAs to annotate. For Verifiability, each HIT consists of a single response with a compensation of \$2.50 per HIT.

B Prompts

C Automatic Evaluation

The prompt used in our experiments is shown in Table 5.

Instructions: In this task, you'll be given an ARTICLE and an extremely simple question-answer (QA), representing a predicate-argument relation. Please indicate whether the QA is supported by the ARTICLE or not, according to the following definition: A QA is supported by the ARTICLE if the meaning of the QA can be deduced from the ARTICLE.

For example, given the article "Real Madrid beats PSG in the UEFA final Champions League": The QA "Who won something? PSG" is not supported because PSG didn't win anything. The QA "What did someone win? The UEFA Champions League" is supported because someone (here Real Madrid) did win the UEFA Champions League.

ARTICLE:

QA:

Is the QA supported by the ARTICLE? Please answer only "Yes" or "No".

Table 5: Our prompt for automatically verifying whether a QASem QA is supported by the reference text, **brown** indicates an example.

¹¹<https://vuejs.org>.